

MTH6134 Practical Week 2

1 The effect of salt on blood pressure

The file `saltBP.txt` contains measurements on 25 elderly people. The variables measured are

- BP – systolic blood pressure
- salt – average daily intake of salt in grams.

Download and save the data set into your R directory. Read in the data set using

```
saltBP_data <- read.table("saltBP.txt", header = T)
```

Draw a scatterplot of BP against salt using `plot(saltBP_data$salt, saltBP_data$BP)`. Is a linear regression model suitable here?

Answer:

A linear regression model seems suitable since there appears to be a linear relationship between the two variables with the points scattered around a straight line. Fit a linear regression model where BP is the dependent variable and salt is the independent variable and look at the summary of the fitted model using

```
saltBP_fit <- lm(BP ~ salt, data = saltBP_data)
summary(saltBP_fit)
```

What is the fitted regression model?

Answer:

$\text{Expected(BP)} = 128.616 + 1.197 \times \text{Salt}$

Use the fitted regression model to answer the following questions:

- What is the effect on blood pressure of increasing salt intake by one gram according to the fitted model?

Answer:

An increase of 1.197

- What is the average blood pressure of someone with a salt intake of six grams per day?

Answer:

$128.616 + 1.197 \cdot 6 = 135.798$

- I currently have nine grams of salt per day and I want to reduce my blood pressure by 5. By how much should I reduce my average daily salt intake?

Answer:

We have

```
x0 <- 9
```

```
y0 <- 128.616 + 1.197*x0
```

And we want

```
y1 <- y0-5
```

Solving $y1 = 128.616 + 1.197 * x1$ with respect to $x1$

```
x1 <- (y1 - 128.616) / 1.197
```

Difference between $x0$ and $x1$ (4.177109g)

```
x0-x1
```

Check the model assumptions by plotting the studentised residuals against the fitted values and using a QQ plot of studentised residuals of the fitted model. Comment on the result.

```
plot(saltBP_fit$fitted, rstudent(saltBP_fit),  
xlab="Fitted values", ylab="Studentised residuals")  
abline(h=0, lwd=2)
```

```
qqnorm(rstudent(saltBP_fit), main=NULL)  
qqline(rstudent(saltBP_fit), lwd=2)
```

Answer:

First plot : the points are randomly scattered and there is no obvious pattern (although one could also argue that there is some periodicity in the point pattern). So no concerns are raised here on the validity of a linear relationship between the response and the explanatory variable or of the assumption of constant variance.

Second plot: the points mostly follow the straight line so no concern here that the assumption of normality is violated.

2 Children's heights and weights

The data set includes 30 data points on children's weights and heights.

Download and save the data set into your R directory. Read in the data set using

```
chw_data <- read.table("heightweight.txt", header = T)
```

Draw a scatterplot of Height against Weight using `plot(chw_data$Height, chw_data$Weight)`. Is a linear regression model suitable here?

Answer:

There is a positive correlation between the two variables but weight appears to have been categorised into six categories and the spread of values for weight within each level of height is increasing as height increases.

Fit a linear regression model where Weight is the dependent variable and Height is the independent variable and look at the summary of the fitted model using

```
chw_fit <- lm(Weight ~ Height, data = chw_data)
summary(chw_fit)
```

Check the model assumptions by plotting the studentised residuals against the fitted values and using a QQ plot of studentised residuals of the fitted model. Plot Cook's distances and look for influential observations. Comment on the result.

```
plot(chw_fit$fitted, rstudent(chw_fit), xlab="Fitted values",
     ylab="Studentised residuals")
abline(h=0, lwd=2)
which_large <- (1:nrow(chw_data))[abs(rstudent(chw_fit))>3]
text(chw_fit$fitted[which_large], rstudent(chw_fit)[which_large],
     labels = (1:nrow(chw_data))[which_large])

qqnorm(rstudent(chw_fit), main=NULL)
qqline(rstudent(chw_fit), lwd=2)

plot(chw_fit, 4)
abline(h=8/(30-2*2), lty=2)
```

Answer:

First plot: there is a funnel shape, with the spread of the residuals increasing as the fitted value increase.

This is not surprising given the scatterplot and therefore the assumption of constant variance is violated here.

Second plot: the points mostly follow the straight line so no concern here that the assumption of normality is violated (although the points on the tails deviate from the straight line).

Third plot: three observations have been labelled as potentially influential (25, 26, 30) but only observation number 30 is above the threshold ($8/(30 - 2 \cdot 2) = 0.31$).

The next step is then to consider a Box-Cox transformation

```
library(MASS)
boxcox(chw_fit, lambda = seq(-30, 30, by = 0.01))
```

What value of λ would you choose here?

Answer:

The profile log-likelihood for λ includes values between (roughly) -5 and -15, and these are all equivalent in terms of the ease of interpretation, so we can choose $\lambda = -10$, which is the closest to the maximum.

Refit the model after transforming your response, check the diagnostic plots and summarise your findings.

```
chw_bc_fit <- lm(1/Weight^(10) ~ Height, data = chw_data)
summary(chw_bc_fit)

plot(chw_bc_fit$fitted, rstudent(chw_bc_fit), xlab="Fitted values",
      ylab="Studentised residuals")
abline(h=0, lwd=2)
which_large <- (1:nrow(chw_data))[abs(rstudent(chw_bc_fit))>3]
text(chw_bc_fit$fitted[which_large], rstudent(chw_bc_fit)[which_large],
     labels = (1:nrow(chw_data))[which_large])

qqnorm(rstudent(chw_bc_fit), main=NULL)
qqline(rstudent(chw_bc_fit), lwd=2)

plot(chw_data$Height, (1/chw_data$Weight^10))
abline(chw_bc_fit)

plot(chw_bc_fit, 4)
abline(h=8/(30-2*2), lty=2)
```

Answer :

Our response is now $1/\text{Weight}^{10}$, the diagnostic plots do not raise any concerns and no observations have Cook's distances greater than 0.3, although we may need to inspect observation 26 a bit more closely.

The summary of the model suggests, as expected, that as the explanatory variable increases, the response decreases (estimated slope is negative), so on the original scale, as height increases, weight also increases (although the relationship between height and weight is not linear, as shown by the fitted lines below).

3 Psychological perception

In this experiment, 24 blindfolded subjects were asked to estimate the direction of the shortest detour line to an un-anchored origin. The data include information on the signed errors in judgements of angular position (Error), the True angle as well as the Path that participants took (1 = clockwise, 2 = counter).

Download and save the data set into your R directory. Read in the data set using

```
errors_data <- read.table("errors.txt", header = T)
```

Attach the data set so that you can refer to the columns without having to call the name of the data set each time `attach(errors_data)`.

Fit a normal linear model, with Error as the response, including the main effects and the interaction between True and Path. Be careful to declare Path as a factor, otherwise R will interpret the 2 to mean twice as high as the 1. Perform diagnostic checks to test the model assumptions. Once you are satisfied that the assumptions hold, construct the anova table and use it to decide on whether you should drop the interaction term from the model.

Answer:

```
full <- lm(Error ~ True * as.factor(Path), data = errors_data)
summary(full)
```

```
plot(full$fitted, rstudent(full), xlab="Fitted values",
     ylab="Studentised residuals")
abline(h=0, lwd=2)
```

```
qqnorm(rstudent(full), main=NULL)
qqline(rstudent(full), lwd=2)
```

```
anova(full)
```

The diagnostic plots do not raise any concerns. The anova table suggests that we can drop the interaction effect from the model (p-value for testing H_0 : interaction is not significant is 0.6417).

Drop the interaction terms and refit the model. Look at the summary of the additive model and comment on the result.

Answer:

```
add <- update(full, .~. - True:as.factor(Path))
```

```
summary(add)
```

Both main effects are significant according to the t-test. We see that True has a negative effect while Path=2 has a positive effect compared to Path=1.

Plot the fitted lines (one for each level of Path) and add confidence intervals and prediction intervals in each case. Comment on the results.

Answer:

```
# Plot points
```

```
plot(True, Error,
      xlim = range(True) + c(-5, 5),
      ylim = range(Error) + c(-5, 5),
      xlab = "True", ylab = "Error", col = c("black", "red")[Path])
```

```
# calculate confidence and prediction intervals
```

```
conf_int <- predict(add, interval = "confidence")
```

```
pred_int <- predict(add, interval = "prediction")
```

```
# fitted lines
```

```
matlines(True[Path==1], conf_int[Path==1, 1],
          lty = 1, col = 1)
```

```
matlines(True[Path==2], conf_int[Path==2, 1],
          lty = 1, col = 2)
```

```
# Confidence intervals
```

```
matlines(True[Path==1], conf_int[Path==1, 2:3],
          lty = 2, col = 1)
```

```
matlines(True[Path==2], conf_int[Path==2, 2:3],
         lty = 2, col = 2)

# Prediction intervals
matlines(True[Path==1], pred_int[Path==1, 2:3],
         lty = 3, col = 1)

matlines(True[Path==2], pred_int[Path==2, 2:3],
         lty = 3, col = 2)
```

Both fitted lines have a negative slope, since, as mentioned above, True has a negative effect on Error. The points are described well by the fitted line for each level of Path and indeed there is no indication that the lines should have different slopes.

The confidence intervals around the fitted lines are, as expected, narrower than the corresponding prediction intervals. There are a few points that are not within the confidence intervals and one that is not in the prediction interval (red).

The values for True seem to be quite separated for the two different levels of Path.

If you are interested, you can read up on the experiment in the original paper

<https://pdfs.semanticscholar.org/8fa1/863ad7006e2e3ffa09e8d256cd24aa2ccae1.pdf>.